

Data transform

What is?

raw data

→ desired form

Why?

reducing skewness
removing outliers
comparing variables on a scale
improve model performance

When?

before modeling

use any algorithm

interpretability ✓

that needs assumption

How?

Types

①

Scaling

Z-score

min-max

Robust

$X - \mu_{\text{median}}$
IQR

$$\begin{aligned} E\left(\frac{X - \mu}{\sigma}\right) &= \frac{1}{\sigma} E(X - \mu) \\ &= \frac{1}{\sigma} [E(X) - \mu] \\ &= \frac{1}{\sigma} (\mu - \mu) = 0 \end{aligned}$$

Z Φ

$$\frac{X - \mu}{\sigma}$$

$$\begin{aligned} \mu &= \frac{\sum x_i}{n} \\ \sigma &= \sqrt{\frac{\sum (x_i - \mu)^2}{n}} \\ s &= \sqrt{\frac{\sum (x_i - \mu)^2}{n-1}} \end{aligned}$$

$$\frac{x_{\text{max}} - x_{\text{min}}}{x_{\text{max}} + x_{\text{min}}}$$

